culations). From the performance point of view, the integer version is better. Here, the firmware (.bin) required is nodemcu_integer_0.9.6-dev_20150627.

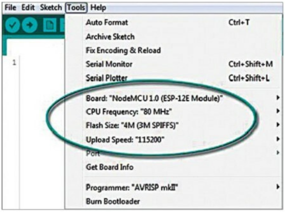

Fig. 5: NodeMCU details on Arduino IDE

The steps to flash the firmware using NodeMCU Flasher are:

1. Connect the board to the computer through USB interface

2. Open NodeMCU Flasher. In 'Config' menu, add the nodemcu bin file on 0x00000 column (refer Fig. 6(a) and (b))

3. Press 'Flash' button under 'Operation' menu (Fig. 6(b)), and wait

After a few seconds, the flashing process should complete. From my experience, you will need to reboot NodeMCU board in order to connect it to the PC. To resolve connection problem, you can unplug and then plug in NodeMCU to the PC again, or press reset (RST) button on NodeMCU board.

(a) (b)

Fig. 6: NodeMCU firmware flashing process: (a) In 'Config' menu of NodeMCU Flasher, add the NodeMCU bin file on 0x00000 column; (b) Press 'Flash' button under 'Operation' menu

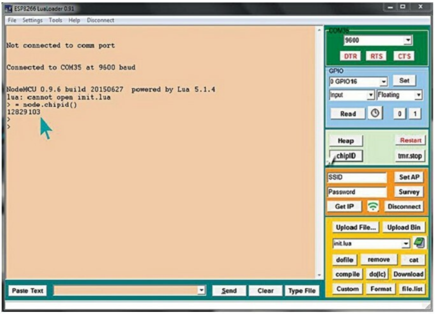

Fig. 7: LuaLoder displays unique chip ID of the connected board

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

NodeMCU and Lua development

The architecture of NodeMCU and availability of many Lua IDEs allow it to be programmed independently, without the need of other microcontrollers like Arduino, for example. I've found a simple IDE called LuaLoder that can be used for NodeMCU Lua development (https://github.com/GeoNomad/LuaLoader). LuaLoder allows you to upload Lua files, execute them, and also interact with the board without undue difficulties. The screenshot of LuaLoder window is shown in Fig. 7.

Another IDE is ESPlorer (*https://github.com/4refr0nt/ESPlorer*). Actually, these IDEs are for uploading files to the ESP8266 and working with the Lua serial interface. They are compatible with almost all versions of Windows, with terminal windows to show the output from the ESP8266 UART.

Terminal windows also let you type or paste commands for immediate interpretation and execution. A selection of buttons is available to automatically type frequently used commands and to select files for uploading to the ESP8266 file system.

At first I got a bit scared by the complex Lua structures. Later I enjoyed playing with NodeMCU and it has been working great for about a month with only a few hiccups. Well, good luck for all your projects! EFY


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